

ALAN LO

Senior Backend Engineer — Distributed Systems & AI Infrastructure

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SUMMARY

Backend and distributed systems engineer with 6+ years of production experience building high-throughput, fault-tolerant systems, extending that foundation into production AI systems and agentic infrastructure through hands-on projects.

EXPERIENCE

Weedmaps | Backend Software Engineer II

Aug 2023 – Feb 2026

Internal Ad Marketplace platform connecting cannabis dispensaries and partnered brands with Weedmaps' e-commerce user base

- **Joined a greenfield, undocumented Ad Marketplace codebase** (AWS EKS, thousands of RPS, 99.9% uptime) alongside one senior engineer, using the codebase itself as the source of truth; wrote the Technical Requirements Document and co-implemented Sponsored Products Ads with targeting across sales region, product, category, and brand, shipping to production with zero downtime.
- **Independently proposed, designed, and delivered** a Redis Pub/Sub cache invalidation architecture, replacing a 3-minute scheduled rebuild with under-30-second real-time propagation across a distributed ~60-pod fleet, eliminating user-facing data inconsistencies; instrumented DataDog metrics to monitor invalidation latency in production.
- **Co-built auction decisioning logic** governing House Ads vs. Paid Ad Inventory, integrating OpenSearch to query live product and inventory data for real-time eligibility decisions.
- **Led the Java 17 → 21 migration** on the same high-traffic microservice: upgraded libraries, replaced deprecated methods, validated through unit/integration tests and load testing in non-production environments, and shipped via canary deployment with zero downtime.
- **Mentored engineers above my own level** (SE3/SSE) on schema migration practices, PR and code review etiquette, and deployment process while titled SE2; tracked feature vs. non-feature work for SOX compliance auditing.

Dish Network | Software Engineer II

June 2019 – July 2023

On-demand video streaming platform serving millions of subscribers — contributed across two teams: Media Platform and Content / On-Demand

- **Media Platform team:** Contributed to a greenfield migration from legacy set-top box hardware to Android; implemented FFmpeg/MPEG-DASH video pipelines and GStreamer-based ad insertion logic consuming Gracenote ad metadata.
- **Content / On-Demand team:** Partnered on Kafka and Spring Boot ingestion pipelines and Elasticsearch indexing for video metadata across partner content (including HBO and Nat Geo); built internal business tooling for content recommendation publishing on Dish set-top box software.
- **Mentored junior and senior engineers** to prepare the team for the greenfield Android set-top box migration, and separately mentored senior engineers on infrastructure and process during On-Demand Content ingestion work.

PERSONAL PROJECTS

- **OpenClaw — Multi-Agent AI Server:** Built and operate a multi-agent AI system on AWS EC2 orchestrating Grok-based agents through a RAG pipeline ingesting SEC filings, HubSpot data, and government records for investment research; designed under a deliberate low-cost compute constraint.

ETH Stablecoin Simulator (*backend verified, frontend in progress*) — Java 25, Spring Boot 4.1, Spring AI, Web3j, PostgreSQL (CQRS/Flyway), Next.js 16 / React 19, Docker

- **Built financial-grade numerical correctness** into a DeFi fixed-income simulator by making Java the single source of truth for all debt, risk, rate, and yield calculations, using BigDecimal with a shared MathContext to eliminate floating-point error across the entire money path, enforced by 181 automated tests and coverage gates.
- **Ensured live financial data never silently fails or corrupts** by building a fault-isolated on-chain ingestion pipeline across 5 protocol adapters (Chainlink, Sky, Liquity, Aave, sGHO) using CQRS, single-writer coordination, and reorg replay.
- **Designed a least-authority LLM integration** that prevents hallucinated financial figures by treating model, prompt, and tool output as untrusted: read-only allowlisted tools, per-request token/cost budgets, and a numeric-authority validator rejecting any AI-generated number without an exact match to a verified backend result.

SKILLS

Languages: Java, Python, C/C++, TypeScript

Backend & Frameworks: Spring Boot, Spring AI, Microservices Architecture, REST API design

AI / ML: RAG Systems, Agentic AI / Multi-Agent Orchestration, LLM Integration (Grok), Prompt Engineering

AI-Augmented Development: Claude Code, GitHub Copilot

Data & Caching: Redis (Pub/Sub), OpenSearch, Elasticsearch, PostgreSQL, Amazon Kinesis, Firehose, Glue Jobs, Athena, Kafka

Cloud & Infrastructure: AWS (EKS, EC2, RDS, S3), Kubernetes, Docker, ArgoCD (CI/CD, GitOps), DataDog

EDUCATION

University of California, Irvine

June 2018

B.S. Electrical Engineering, Specialization in Digital Signal Processing

Relevant Coursework: Data Structures & Algorithms, Machine Vision, DSP Processor Lab & Design, Object-Oriented Systems

DeepLearning.ai Specializations: RAG Systems, AI Agents, PyTorch, Machine Learning

Languages Spoken: Spanish (Native), Cantonese (Native), Mandarin (Limited)